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SUPERSEDING
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MILITARY SPECIFICATION

PRINTER, PROJECTION, PHOTOGRAPHIC EN-36( )

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of photographic projection printer, designated as Printer, Projection, Photographic EN-36( ) (see 6.6).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
QQ-S-781	Strapping, Steel, Flat and Seals
NNN-P-40	Paper, Lens
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-C-843	Cushioning Material, Cellulosic
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Ft. Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6740

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MILITARY

MIL-F-32	Film, Photographic, Aerial, Black and White
MIL-P-116	Preservation-Packaging, Methods of
MIL-B-117	Bag, Sleeve and Tubing-Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-675	Coating of Glass Optical Elements (Antireflection)
MIL-T-713	Twine, Impregnated, Lacing and Tying
MIL-I-3158	Insulation Tape, Electrical Glass-Fiber (Resin-filled) and Cord, Fibrous-glass
MIL-D-4826	Developer, Photographic, Low Contrast, Armed Forces Developer No 3.
MIL-E-6060	Envelope, Packaging, Water Vapor-proof Flexible
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Electronic Equipment
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)
MIL-O-16898	Optical Elements, Packaging of
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-S-23190	Strap, Tie Down, Electrical Components
MIL-I-24391	Insulation Tape, Electrical, Plastic Pressure Sensitive
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, For Packaging
MIL-T-43435	Tape, Lacing and Tying

STANDARDS

FEDERAL

FED STD 356	Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-150	Photographic Lenses
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment

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MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Charcteristics, Measurement of
MIL-STD-731	Quality of Wood Members for Containers and Pallets
MIL-STD-810	Environmental Test Methods

DRAWINGS

US ARMY ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
DL-SC-B-664651	Printer, Projection, Photographic EN-36B

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2(b)) the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. Printer, Projection, Photographic EN-36( ) shall be constructed in accordance with Drawing and Data List DL-SC-B-664651.

3.3 Cleaning. Cleaning of materials, parts and equipments shall be in accordance with requirement 9 of MIL-STD-454 and the following: Materials, parts and equipment shall be cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, solder, flux residues, outgassing products and any other contaminant. The cleaning procedure and cleaning agents shall have no deleterious effect on the performance requirements of the materials, parts and equipments. The same cleaning procedures shall be applied to all equipment produced and delivered on a contract.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings (see 4.4).

3.5 Marking.

3.5.1 General. Marking shall conform to MIL-M-13231 (see 4.4). Front panel marking shall be Group I as described in that specification.

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3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Each Printer, Projection, Photographic EN-36( ) shall be serial numbered.

3.6 Coating of lenses. All glass-to-air surfaces of the lenses shall be coated in accordance with MIL-C-675 to reduce reflection losses at each surface.

3.7 Riveting. Riveting shall conform to requirement 12 of MIL-STD-454.

3.8 Soldering. Soldering shall conform to requirement 5 of MIL-STD-454, except as specified herein. The liquid flux shall be type R or RMA of MIL-F-14256 and flux-cored solder shall be type R or RMA of QQ-S-571.

3.9 Wiring and cabling. Wiring and cabling shall be neat and sturdy.

3.9.1 Cabling. Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected there by or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by lacing or tying tape conforming to MIL-T-43435, twine type P nylon conforming to MIL-T-713, pressure sensitive tape conforming to MIL-I-24391, or straps for cable harnessing conforming to MIL-S-23190; however, cord conforming to MIL-I-3158 shall be used where resistance to heat or chemicals is essential. Individual conductors thus secured shall lie essentially parallel; however this does not prohibit the use of twisted pairs.

3.9.2 Slack. Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting the parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected.

d. To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement.

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e. To facilitate field repair of broken or cut wires.

f. To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts.

3.9.3 Protection. Wires and cables shall be so placed and protected as to avoid contact, under equipment service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal partitions, shields, or similar items less than 3.2mm (1/8 inch) in thickness, the wires shall be protected by suitable grommets or bushings. Where wires run through holes in metal 3.2mm (1/8 inch) in thickness or greater, the wires shall be protected by grommets or bushings or shall have the edges of the holes rounded to a minimum radius of 1.6mm (1/16 inch).

3.9.4 Support. Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions, or (ii) after service or repair of the equipment in a normal manner. Twine or tape shall not be used for securing wire and cable.

3.9.5 Clearance. Clearance between wires or cables, and parts, such as resistors, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to equipment service conditions. Clearance between solder connection or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to equipment service conditions.

3.9.6 Splicing and stretching. Wires in a continuous run between two terminals shall not be spliced or stretched during the cabling and wiring operation.

3.10 Input voltage and frequency. Printer, Projection, Photographic EN-36( ) shall be capable of operating from a 115 volts ±10 volts, 47.5 to 63.0 hertz AC source.

3.11 Light-tightness. Variable contrast photographic paper shall not be fogged under normal handling conditions from the projection head assembly or an ambient light level of 107.6 to 161.5 Lux (10 to 15 foot candles) around the outside of the equipment (see 4.16).

3.12 Optical system operation.

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3.12.1 Distortion. The curvature of projected images of the lines adjacent to both the long and short edges of the picture aperture shall be not more than 0.5 percent of their length (see 4.7).

3.12.2 Resolving power (see 4.21).

3.12.2.1 90mm lens. The minimum resolving power of the 90mm lens, with the diaphragm set at f/6.3 opening, shall be not less than 40 lines per millimeter at the axis and 30 lines per millimeter at any other point in the field.

3.12.2.2 127mm lens. The minimum resolving power of the 127mm lens, with the diaphragm set at f/6.3 opening, shall be not less than 32 lines per millimeter at the axis and 24 lines per millimeter at any other point in the field.

3.12.3 Lateral color. The projection lens shall not have lateral color in excess of 0.06 percent of the semi-diagonal of the field for any wave-length of light between 440 and 650 millimicrons (see 4.14).

3.12.4 Longitudianal color (see 4.17). In order to meet the longitudinal color requirement, the printer shall be capable of meeting the requirements specified for resolving power in 3.12.2 with filters having dominant wave-lengths and transmission bands as follows installed in the light beam:

650 millimicrons (Transmission of wave-lengths shorter than 5 percent millimicrons should be negligible)

440 millimicrons (370 to 510 millimicrons)

3.12.5 Magnification. Printer, Projection, Photographic EN-36( ) shall be capable of 4X magnification of the 70mm film frame and 2X magnification of the 127mm (5 inch) film frame (see 4.18).

3.12.6 Uniformity of illumination. At maximum magnification, the illumination of the projected image shall be free from objectionable bands or patches differing in color or brightness from the adjacent parts of the illuminated area. The average illumination at the four corners shall be not less than 65 percent of the illumination at the center. At no corner shall the illumination be less than 50 percent of the illumination at the center (see 4.24).

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3.12.7 Exposure index control. The printer exposure index control system shall be capable of printing neutral densities from 0.6 to 1.8 (representing average negative densities of 0.6 to 1.8) and controlling the level of exposure so that any of these densities print to essentially the same print density ±0.1. The above tolerance is applicable for any one printing paper using a 1.0 ±0.5 neutral density as a negative to obtain a reflected print density of .63 ±0.5 (see 4.10).

3.13 Safety filter. The transmission of the filter shall be such that no fogging of variable contrast photographic paper will take place after exposure to the filtered light beam for a period of at least 5 minutes (see 4.22).

3.14 Paper drive mechanism. The drive mechanism shall be capable of accommodating 238.1mm (9-3/8-inch) wide roll printing paper in lengths of 61 meters (200 feet). The drive mechanism shall maintain the paper under constant and uniform tension at all times and shall automatically advance the paper in accordance with the required format 228.6mm x 279.4mm (9 inch x 11 inch) prints from 70mm film and 228.6mm x 228.6mm (9 inch x 9 inch) prints from 127mm (5 inch) film (see 4.20).

3.14.1 Alarm. When supply of paper is exhausted in the supply magazine, an alarm shall signal the operator both audibly and visually.

3.15 Film transport. Printer, Projection, Photographic EN-36( ) shall be capable of accommodating perforated 70mm roll film and 127mm (5 inch) roll film in lengths up to 61 meters (200 feet) on standard spools. The film mechanism shall be capable of moving either film widths, one frame at a time, smoothly past the projection head aperture without any deleterious effects to the film, such as scratching, tearing, etc. The film shall be accurately positioned at the focal plane (see 4.11).

3.16 Film temperature rise. The temperature of the negative in the film gate shall not rise more than 60 degrees above the ambient temperature (see 4.12).

3.17 Service conditions. The equipment shall meet the following service conditions:

3.17.1 Temperature.

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3.17.1.1 Low. Before, during, and after testing as specified in 4.23.1, the equipment shall meet full specified requirements as specified in 3.11 through 3.15.

3.17.1.2 High. Before, during, and after testing as specified in 4.23.2, the equipment shall meet full specified requirements as specified in 3.11 through 3.16.

3.17.2 Humidity. Printer, Projection, Photographic EN-36( ) shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials after being tested as specified in 4.19.

3.17.3 Altitude (elevation). The equipment shall meet full specification performance as specified in 3.11 through 3.15 after being tested as specified in 4.9.

3.17.4 Bounce. When specified in the invitation for bids and contract (see 6.2(c)), the equipment shall show no physical damage and shall be capable of meeting full specification requirements after being tested as specified in 4.6.

3.17.5 Vibration. After testing as specified in 4.25, the printer shall be operable without degradation in specified performance and shall sustain no physical damage.

3.18 Electromagnetic interference characteristics. The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461, Table A-II MIL-I-11748 for Class II B. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification (see 4.8).

3.19 Life. When tested as specified in 4.15, Printer, Projection, Photographic EN-36( ) shall operate without failure. A burned-out projection lamp shall not be considered a failure.

3.20 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.13).

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3.21 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs herein:

3.3 Cleaning
3.7 Riveting
3.8 Soldering
3.9.2 Slack in wires and cables
3.9.3 Protection for wires and cables
3.9.5 Clearance in wiring and cabling
3.9.6 Splicing and stretching of wires

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Packaging inspection (see 4.27).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish two first article units of Printer, Projection, Photographic EN-36( ).

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4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4 and the inspections specified for Group A, Group B, and Group C (See Table I, II and III, respectively). The inspection shall be performed in the following order: 4.4 Group A and Group B for all units, and designated units for specific subgroups within Group C in accordance with 4.5.3.1. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspections.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Finish, protective	3.4
Marking	3.5
Coating of Lens	3.6
Riveting	3.7
Soldering	3.8
Cabling	3.9.1

4.5 Quality conformance inspection. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table I.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

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Table I. Group A Inspection

Inspection	Req Para	Test Para	AQL Major	AQL Minor
Visual and Mechanical	3.21	4.26	1.0%	4.0%
Operational				
Distortion	3.12.1	4.7		
Exposure Index Control	3.12.7	4.10		
Film Transport	3.15	4.11	1.0%	*
Light Tightness	3.11	4.16	For the	
Paper Transport System	3.14	4.20	Group	

*All operational defects are considered major

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

Table II. Group B Inspection

Inspection	Req Para	Test Para	AQL
Interchangeability	3.20	4.13	6.5%
Lateral Color	3.12.3	4.14	
Longitudinal Color	3.12.4	4.17	
Magnification	3.12.5	4.18	4.0%
Resolution	3.12.2	4.21	For the
Uniformity of Illumination	3.12.6	4.24	Group

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL's listed in Table II, shall be as follows:

AQL	Inspection Level
4.0 Percent	S-4
6.5 Percent	S-4

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4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. One sample unit shall be selected at random for each sub-group of inspections in Table III from every fifty units, or fraction thereof, produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

Table III. Group C Inspection

Inspection	Req Para	Insp Para	Order of Insp
Subgroup 1			
Electromagnetic Interference	3.18	4.8	1
Life	3.19	4.15	2
Subgroup 2			
Film temperature rise	3.16	4.12	1
Low temperature	3.17.1.1	4.23.1	2
High temperature	3.17.1.2	4.23.2	3
Elevation	3.17.3	4.9	4
Humidity	3.17.2	4.19	5
Vibration	3.17.5	4.25	6
Bounce (If specified, see 6.2)	3.17.4	4.6	7

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in accordance with Table III.

4.5.3.3 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2(k)).

4.5.4 Reinspection of conforming Group B sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or purchase order, provided sample units are resubjected to and pass Group A inspection after all damage is repaired, except sample units subjected to the life test in Table III shall not be delivered.

4.6 Bounce test. The equipment, packaged for shipment, shall be tested in accordance with Method 514, Procedure XI of MIL-STD-810. Prior to and after testing, the equipment shall meet full specification performance. When practicable, package tester accessories shall be selected from those listed on SC-A-46439.

4.7 Distortion. The image of a test plate, having straight lines adjacent to the four edges of the film aperture, shall be projected onto a smooth white paper surface on an easel. The projected image of the lines shall be compared with straight lines joining opposite ends of the projected image lines on the easel, and shall not be distorted more than that specified in 3.12.1.

4.8 Electromagnetic interference test. The equipment shall be tested for electromagnetic interference characteristics in accordance with MIL-STD-462.

4.9 Altitude. The equipment shall be subjected to the test of MIL-STD-810, Method 500.1, Procedure I. The test procedure shall include a 40,000 feet altitude non-operational exposure for not less than one hour previous to the 15000 feet altitude operational portion of step 2. Altitude rate changes should not exceed 2000 feet per minute.

4.10 Exposure index control. The exposure index control shall be tested to determine compliance with the requirements specified in 3.12.7.

4.11 Film transport test. The film carriers shall be loaded with both sizes of exposed processed film, 70mm and 5 inch, and placed alternately in the equipment. The film shall be moved passed the projection head aperture. The film shall be inspected, using a 20 power magnifier, for any evidence of scratching or other defects (see 3.15).

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4.12 Film temperature rise. A test film having a uniform density of 2.0 or greater shall be placed in the film carrier. The printer shall be operated intermittently for a period of 60 minutes, cycled 10 seconds on and 10 seconds off. At the end of this period, the temperature on the face of the negative shall be determined by the use of a suitable thermocouple of 0.100mm (0.00396") or 0.089mm (0.00353 inch) diameter wire (No 38 or 39 AWG), which should be held or cemented to the emulsion side of the test film at the center of the aperture. The temperature rise shall be determined with an instrument accurate to ±2.2 degrees C (±4 degrees F). The test film shall be placed in the film carrier with the emulsion side away from the light source (see 3.16).

4.13 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.20. When a listed dimension is not within specified limits, it shall be considered a major defect.

a. Mating dimensions of the film carrier and printer.

b. Mating dimensions of the lens and the projection head.

c. Mating dimensions of the paper supply compartment and take-up magazine.

4.14 Lateral color. Printer, Projection, Photographic EN-36( ) shall be tested using Method 20 of MIL-STD-150 to determine compliance with the requirement specified in 3.12.3.

4.15 Life. Printer, Projection, Photographic EN-36( ) shall be operated for 150,000 cycles. A complete cycle shall be 2 seconds. A photographic print shall be made after every 5000 cycles using the negative specified in 4.21 and a sheet of variable contrast photographic paper. The print shall be checked for photographic quality (see 3.19).

4.16 Light-tightness. A short length of variable contrast photographic paper shall be loaded into the paper drawer compartment and the unit placed in the equipment. Cap the lens and turn the projection lamp on for a period of at least 2 minutes. The light level around the outside of the equipment shall be 107.6 to 161.5 Lux (10 to 15 foot candles). The short length of photographic paper shall be processed along with a sheet of variable contrast photographic paper which has been kept in total darkness and then examined for any difference in density using a standard laboratory densitometer (see 3.11).

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4.17 Longitudinal color. Using the filters specified in 3.12.4, Printer, Projection, Photographic EN-36( ) shall be tested in accordance with Method 18 of MIL-STD-150.

4.18 Magnification. The projection printer shall be tested to determine compliance with the requirement specified in 3.12.5.

4.19 Humidity. The equipment shall be subjected to the test of MIL-STD-810, Method 507.1, Procedure II with the exception that step 6 shall be limited to two 48 hour cycles. Operation and measurements shall only be performed during the last 5 hours of the last cycle.

4.20 Paper transport system. The paper supply compartment and take-up magazine shall be loaded with a 15.2 meters (50 feet) dummy roll of paper. The indicator shall be checked. While transporting the paper from the supply compartment to the take-up magazine, the positive stops for 9 and 11 inch (228.6 and 279.4 millimeters) print size shall be checked for operation. The entire roll of paper shall be passed through the transport system and the alarm specified in 3.14.1 shall be checked for operation.

4.21 Resolution.

4.21.1 Test negative. The resolution test negative, supplied by the contractor, shall contain high contrast target positioned in the format as referenced in the figure titled "Projected Resolving Power Test Plate" of MIL-STD-150. The target patterns on the test negative shall cover a range of at least 10 to 110 lines per millimeter. The dark area of the test patterns shall have a photographic density which is at least 2.0 greater than the density of the light area. The individual targets shall be made up darklines on a light background.

4.21.2 Test method. The lens shall be tested in accordance with Method 14 of MIL-STD-150 with the following exception: The photosensitive material shall be duplicating film in accordance with MIL-F-32. The film shall be developed using Developer No 3, per MIL-D-4826, for four minutes at +20 degrees C (+68 degrees F). Each lens shall be tested at the specified aperture opening with no contrast filter being used. The resolution shall not be less than that specified in 3.12.2.

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4.22 Safety filter. With a sheet of high speed variable contrast photographic paper placed on the printing area of the base and the safety filter placed in the light beam path, the projection lamp shall be turned on for at least five minutes. After the sheet of photographic paper has been developed along with another sheet of photographic paper, which has been kept in total darkness, the sheets shall be examined for any difference in density using a standard laboratory densitometer (see 3.13).

4.23 Temperature test.

4.23.1 Low temperature. The equipment shall be subjected to the test of MIL-STD-810, Method 502.1, Procedure I. The temperature in Step 2 shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (step 4) shall be 4.4 degrees C (+40 degrees F).

4.23.2 High temperature. The equipment shall be subjected to the test of MIL-STD-810, Method 501.1, Procedure II. The temperature of Step 7 shall be 46.1 degrees C (115 degrees F).

4.24 Uniformity of illumination. The projection printer shall be set at its maximum enlargement ratio; the lens set of its largest aperture; and the lens focused to give a sharp image of the negative aperture formed by a clear glass glide masked for maximum picture size. The printing area shall be covered with a sheet of white matte paper over the illuminated area. Photometric measurements shall be made at the center and each of the four corners. Corner illumination shall be measured as close to the extreme corners of the field as possible. The presence of objectionable bands or patches differing in color or brightness from the adjacent parts of the illuminated area, shall be noted. The illumination shall be as specified in 3.12.6.

4.25 Vibration. The equipment shall be subjected to sinusoidal vibration having a double amplitude of .762mm (0.030 inch), the frequency being varied uniformly between the limits of 10 Hz and 55Hz. The rate of frequency change shall be linear and shall be such that a complete cycle (10 to 55 to 10 Hz) shall be traversed in 30 minutes. One cycle shall be performed along each of the three major axes of the equipment for a total test time of 90 minutes. At the completion of the test, the equipment shall meet specified performance requirements and have no physical damage.

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4.26 Visual and mechanical inspection. Printer, Projection; Photographic EN-36( ) shall be examined for the defects listed in MIL-STD-252.

4.27 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the Group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.28 Rough handling test. When rough handling test in accordance with MIL-P-116 is required by the contract (see 6.2(e)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

4.11 Film transport
4.16 Light-tightness
4.20 Paper transport system

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, B, or Commercial, as specified (see 6.2(d)).

5.1.1 Level A.

5.1.1.1 Cleaning. The projection printer shall be cleaned in accordance with process C-1 of MIL-P-116. Lenses shall be cleaned in accordance with MIL-O-16898.

5.1.1.2 Drying. The projection printer shall be dried in accordance with the applicable procedures of MIL-P-116. Lenses shall be dried in accordance with MIL-O-16898.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged Method IC-1.

5.1.1.4.2 Roll, filter. Each filter roll shall be individually unit packed method 1A8. The heat seal bag shall conform to MIL-B-117, type III, class E, style 1.

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5.1.1.4.3 127MM and 90MM lenses. The 127MM and 90MM lenses shall be removed from the projector and individually unit packed in accordance with MIL-O-16898.

5.1.1.4.4 Printer, Projection, Photographic EN-36( ). Each photographic projection printer shall be individually unit packed method 1A-16 as follows: the assembly shall be securely anchored to the base of one of the containers specified in 5.2 by use of the four integral mounting pads. Equipment shock mounts shall either be immobilized or replaced with shipping shock mounts or other devices. All latches, levers and moveable components shall be adequately secured to prevent accidental opening or free movement. The condenser lens assembly shall be cushioned and blocked by placing polyurethane foam conforming to MIL-P-26514 and 19.1mm (3/4 inch) lumber conforming to MIL-STD-731 between the lens and top plate assembly. The lumber shall be insulated from the top plate assembly with barrier material conforming MIL-B-121, type II, grade A. Lens tissue conforming to NNN-P-40, type II, shall be placed between the clear lens and the polyurethane foam. The power cable assembly shall be coiled to the smallest appropriate diameter and secured with twine conforming to T-T-871, type II, at three equidistant places. The power cable, projection lamp, film stages, film spools and equipment shock mounts (if removed), shall each be individually cushioned with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class domestic, variety SW, grade 275, designed to protect the items from shock and vibration hazards during handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned items, and the items unit packed as specified in 5.1.1.4.2 and 5.1.1.4.3, shall be enclosed within two close-fitting fiberbaord boxes conforming to PPP-B-636, W5c. The boxes shall be closed in accordance with the appendix of the box specification and secured to the top plate assembly, one on either side of the head assembly. Projections, sharp edges or other features of the assembly which may damage the gasketed barrier shall be cushioned with cellulosic wadding conforming to PPP-C-843 or paperboard conforming to PPP-P-291, type II. The cushioned assembly shall be enclosed within a heat sealed bag conforming to MIL-E-6060. The technical literature unit packed as specified in 5.1.1.4.1, shall be secured to the top of the sealed bag.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1, except that method III shall be substituted for method 1A16.

5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

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5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2(d)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. Printer, Projection, Photographic EN-36( ), unit packed as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, overseas type or PPP-B-621, style 4, class 2. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.2 Level B. Printer, Projection, Photographic EN-36( ) unit packed as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, domestic type or PPP-B-621, style 4, class 1. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. Printer, Projection, Photographic EN-36( ) is used for printing 70mm or 127mm (5 inch) aerial camera film negatives in rolls up to 61 meters (200 feet). The printer exposes 238.1mm (9-3/8 inch) wide by 61 meters (200 feet) rolls of variable contrast paper to produce 228.6 x 279.4mm (9 inch by 11 inch) prints from 70mm film and 228.6 x 228.6mm (9 inch by 9 inch) prints from 127mm (5 inch) film.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

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b. When a first article is required for inspection and approval (see 3.1; 4.3 and 6.3).

c. If Bounce requirement is applicable (see 3.17.4).

d. Level A, level B, or Commercial preservation and packing (see Section 5).

e. When the rough handling test is required (see 4.28).

f. Preproduction pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted
Inspection to be performed thereon (see 6.4)

g. Marking and shipping of samples.

h. Place of final inspection.

i. Technical literature required.

j. Quantity of tools and running spare parts required.

k. Group C failure action (see 4.5.3.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample unless otherwise specified. The first article should consist of two units (see 4.3.1). The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Pre-production pack. When specified in the bid request or contract, one acceptable model of Printer, Projection, Photographic EN-36( ), will be packed in a manner that will pass the first article inspection prescribed in MIL-P-116. Illustrations of the packing procedure, together with Bills of Materials, will be prepared and furnished in accordance with the provisions of SC-A-362100.

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6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example EN-36W. The contractor should apply for nomenclature in accordance with the applicable provision in the contract (see 1.1).

6.7 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of the corrective action (see 4.5.3.3).

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-P-55157A(EL) PRINTER, PROJECTION, PHOTOGRAPHIC EN-36( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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